

5 Extra controls

Now assign the keys that the second player will use to control the new caterpillar. The code here uses “w” for up, “a” for left, “s” for down, and “d” for right, but feel free to try out different choices. You’ll need four new functions and four uses of **onkey** to tie the new keys to the new functions.

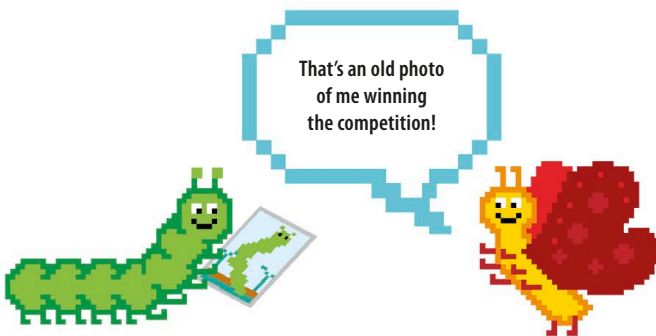
```
def caterpillar2_move_up():
    if caterpillar2.heading() == 0 or caterpillar2.heading() == 180:
        caterpillar2.setheading(90)

def caterpillar2_move_down():
    if caterpillar2.heading() == 0 or caterpillar2.heading() == 180:
        caterpillar2.setheading(270)

def caterpillar2_move_left():
    if caterpillar2.heading() == 90 or caterpillar2.heading() == 270:
        caterpillar2.setheading(180)

def caterpillar2_move_right():
    if caterpillar2.heading() == 90 or caterpillar2.heading() == 270:
        caterpillar2.setheading(0)

t.onkey(caterpillar2_move_up, 'w')
t.onkey(caterpillar2_move_right, 'd')
t.onkey(caterpillar2_move_down, 's')
t.onkey(caterpillar2_move_left, 'a')
```



△ Make it competitive

See if you can figure out how to adapt the two-player game to record each player’s score and then declare the winner at the end. Here’s a tip: you’ll need a new variable to keep track of the second player’s score. When a caterpillar eats a leaf, you’ll need to add a point only to that caterpillar’s score. Finally, when the game is over, you can compare the scores to see who’s won.

▽ Make it harder or easier

If you alter the values inside the loop that increase the length (+1) and speed (+2) of the caterpillar, you can change the difficulty of the game. Higher numbers will make the game harder, while lower numbers will make it easier.

